# Water Purification

# (Household Distilled Water Maker with Least Power – IoT + Mobile App)

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Project Proposal Report

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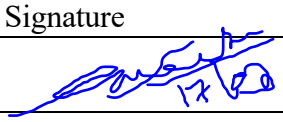
March 2026

## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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## ABSTRACT (Maximum 300 words)

Distilled water, prized for its high purity, is essential for a range of applications, from laboratory experiments and medical equipment to prolonging the life of household appliances like steam irons, humidifiers, and CPAP machines. However, its production via conventional distillation is an energy-intensive process. Traditional countertop distillers often consume between 3-4 kWh to produce one gallon of water, making them costly to operate and environmentally less sustainable for regular household use. Furthermore, these devices typically operate as "dumb" appliances, offering users no insight into their performance, energy consumption, or maintenance needs.

This research proposes the design and development of a Smart, Energy-Efficient Household Distilled Water Maker. The core innovation lies in optimizing the thermodynamic efficiency of the distillation cycle through a combination of passive and active techniques. This includes enhanced insulation, waste heat recovery via a heat exchanger to preheat incoming water, and intelligent, PID (Proportional-Integral-Derivative) controlled heating cycles.

Beyond energy efficiency, the system is transformed into a smart appliance through the integration of Internet of Things (IoT) technology. An ESP32 microcontroller will serve as the system's brain, monitoring key parameters via sensors (temperature, water level, ambient humidity) and controlling the heating element. It will connect to a cloud platform, enabling a custom-developed mobile application to provide users with real-time data visualization, remote start/stop control, push notifications for maintenance (e.g., "cleaning cycle due," "water tank empty"), and detailed analytics on energy consumption and water production. The goal is to create a sustainable, convenient, and intelligent appliance that makes producing high-purity distilled water at home both practical and efficient.

*Keywords: Distilled Water, Water Purification, Household Appliances, Energy Efficiency, IoT (Internet of Things), Mobile Application, Temperature Monitoring, Low-Power System, Smart Water Maker, Remote Control*

# ACKNOWLEDGEMENT

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A special note of appreciation goes to my research partners, **R.M.N.A. Rathnayake**, **W.N.N. De Silva**, **D. Dodanwela** for the collaborative spirit and shared vision that makes integrating our respective components the intelligent "brain" and the autonomous "explorer", a compelling and unified project.

Finally, my heartfelt thanks go to my family and friends for their unwavering encouragement, understanding, and patience throughout the demanding phases of this academic pursuit. Their belief in my work has been a constant source of motivation.

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# LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| **ML** | Machine Learning |
| **IoT** | Internet of Things |
| **TDS** | Total Dissolved Solids (a measure of combined dissolved minerals and salts) |
| **PWM** | Pulse Width Modulation (for controlling heating element power) |
| **ANN** | Artificial Neural Network |
| **SVM** | Support Vector Machine |
| **PID** | Proportional-Integral-Derivative (a control loop mechanism) |
| **ESP32** | A low-cost, low-power system on a chip microcontroller with integrated Wi-Fi |
| **k-NN** | k-Nearest Neighbors |
| **RC** | Radio-Controlled (for manual override, if applicable) |
| **UI/UX** | User Interface / User Experience |
| **API** | Application Programming Interface |
| **JSON** | JavaScript Object Notation (a lightweight data-interchange format) |
| **BOM** | Bill of Materials |

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# 1. INTRODUCTION

## 1.1 Background and Context

Distillation, a process mimicking the natural hydrological cycle, is one of the oldest and most reliable methods for producing high-purity water. By heating water to its boiling point, it separates pure $H_2O$ in the form of steam from virtually all non-volatile contaminants, including heavy metals (lead, arsenic), dissolved salts, minerals, bacteria, viruses, and microplastics. The steam is then cooled and condensed back into a liquid state, yielding water of exceptional purity, typically with a Total Dissolved Solids (TDS) level of 0-5 ppm.

This high purity makes distilled water indispensable in specific contexts:

- **Medical and Laboratory:** For sterilizing equipment, preparing solutions, and in procedures where water purity is critical.
- **Automotive and Appliance Maintenance:** For use in lead-acid batteries, steam irons, humidifiers, and CPAP machines to prevent mineral buildup (scaling) that can damage these devices.
- **Household Consumption:** Some individuals prefer distilled water for drinking, cooking, and preparing baby formulas, believing it to be free from potential contaminants found in tap water.

Despite its effectiveness, the widespread household adoption of distillation is hampered by one major drawback: **high energy consumption**. The latent heat of vaporization (the energy required to turn liquid water into vapor) is substantial. Conventional electric countertop distillers are essentially powerful kettles that run for hours, leading to significant electricity bills and a larger carbon footprint. This energy inefficiency creates a critical barrier, limiting a highly effective purification method to niche applications.

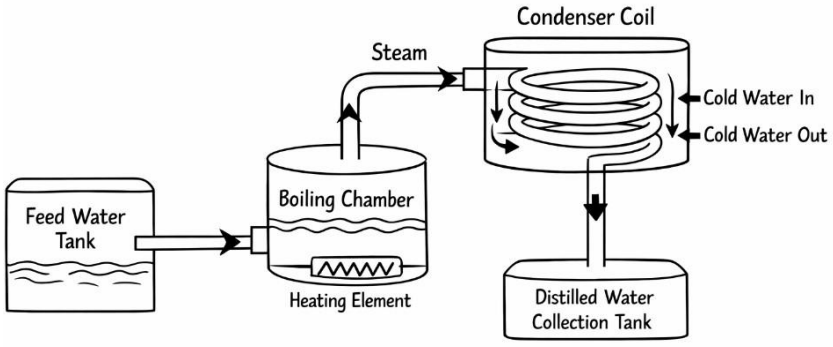



*Figure 1. Basic Principle of Water Distillation*

## 1.2 Literature Review

- **Water Purification Technologies:** A comparative study by Sharma et al. (2019) evaluated various purification methods, including filtration, reverse osmosis (RO), UV treatment, and distillation. The study confirmed that distillation, along with RO, is one of the most effective methods for removing a broad spectrum of contaminants, including dissolved solids. However, it explicitly highlighted distillation's high energy demand as its principal disadvantage compared to other methods.[3]
- **Energy-Efficient Distillation Techniques:** Research into improving distillation efficiency has explored several avenues. Kim and Lee (2018) investigated energy-efficient distillation techniques for household applications, focusing on multi-stage flash distillation and vapor compression, which are common in large-scale industrial plants but are too complex and costly for home use. They also explored passive methods like improved insulation and heat recovery. The use of a **heat exchanger** to preheat the incoming feed water with the waste heat from the condensing steam is a well-established principle for significantly reducing overall energy input.[1]
- **IoT in Water Purification and Home Appliances:** The integration of IoT into home appliances is a growing field. Patel and Mehta (2020) proposed an IoT-based water purification system (primarily for filtration) that used sensors to monitor water quality and transmit data to a mobile app. Singh et al. (2021) provided a broader overview of IoT-based monitoring for home appliances, including water purifiers, demonstrating the feasibility and user benefits of remote monitoring, predictive maintenance alerts, and energy usage tracking. This body of work establishes the technical and user-experience foundation for adding "smart" capabilities to a household appliance.[2][4]




*Figure 2. Block Diagram of Conventional Distiller*

## 1.3 Research Gap

The consumer market for point-of-use water treatment is dominated by filtration (activated carbon, reverse osmosis) and, to a lesser extent, ionization. Distillation, despite being the gold standard for contaminant removal capable of eliminating dissolved solids, heavy metals, bacteria, and viruses with near-total efficacy remains a niche and often overlooked solution for household use. A critical examination of the existing landscape reveals why: commercially available household distillers are technologically stagnant. They are relics of a bygone design era, characterized by crude engineering, energy inefficiency, and a complete lack of user intelligence. While industrial distillation has benefited from decades of optimization in thermodynamics and process control, and while the broader appliance market has been transformed by IoT and smart technology, the household distiller has been left behind.

This research addresses this significant void by proposing the first **intelligent, energy-optimized distillation system** that co-designs hardware efficiency with software intelligence for the consumer market. The gap is not merely incremental but represents a convergence of three distinct failures: thermodynamic, informational, and experiential.

1. **The Thermodynamic Gap: The Unrealized Potential of Energy Efficiency.**

The fundamental barrier to wider adoption of household distillation is its perception and reality as an energy-intensive process. This reputation, however, is not a law of physics but a consequence of poor engineering. Most consumer distillers operate on a simple, brute-force principle: an electrical heating element boils water, and the steam is passively cooled, often with waste heat dumped directly into the room. This represents a profound **thermodynamic gap** between what is theoretically achievable and what is practically implemented.

Industrial-scale distillation systems have long employed sophisticated techniques to maximize efficiency, including:

- **Intelligent Thermal Control:** The use of Proportional-Integral-Derivative (PID) controllers to precisely manage heating elements, minimizing energy overshoot and maintaining optimal boiling temperatures without wasteful cycling.
- **Waste Heat Recovery:** The implementation of heat exchanger systems where the incoming cold feed water is preheated by the outgoing hot distillate and the latent heat of condensation, recovering a substantial portion of the energy input.

These established efficiency principles have not been effectively miniaturized, cost-reduced, and packaged for the consumer market. The gap is not a lack of available technology, but a lack of **applied engineering innovation** that translates industrial best practices into a compact, affordable, and reliable home appliance. A household distiller that could achieve a significant reduction in energy consumption per liter would transform its market viability, making it a sustainable and practical choice for daily use rather than an emergency backup.

2. **The Informational Gap: The Opaque "Black Box" of Operation.**

Even in their current inefficient state, household distillers could offer more value if they were not fundamentally opaque. The second dimension of the gap is the **complete absence of data-driven feedback and user visibility**. Existing devices are "black boxes" that provide no insight into their own operation or performance. A user typically has no way of knowing:

- **Operational Metrics:** How much clean water has been produced in a session? What was the total energy consumed? How long until the current cycle completes?
- **Maintenance Status:** Is scale building up on the heating element? When was the last time the unit was cleaned? When will it next require descaling to prevent damage or efficiency loss?
- **Performance Verification:** Is the unit producing pure water? While distillation is inherently effective, a sudden change in taste or a mechanical issue could compromise output without any alert.

This informational void prevents proactive maintenance, informed usage, and user trust. In an era where consumers expect real-time data from their appliances, from smart refrigerators to robotic vacuums, the household distiller remains stubbornly mute. The gap is the absence of a **sensory and communication layer** that transforms the device from a passive tool into an active, communicative participant in the smart home ecosystem.

**3. The Integration Gap: The Missing Convergence of Hardware and Software.**

The final and most critical dimension is the **systemic failure to integrate optimized hardware with intelligent software**. These two domains are typically treated as separate concerns, but a truly transformative product requires their co-design from the ground up.

An energy-optimized distiller without a smart interface is still a "dumb" appliance it may be efficient, but it cannot be managed or monitored. Conversely, a smart monitoring system bolted onto an inefficient distiller merely provides data about wasteful processes. The true gap is the absence of a **holistic, co-designed system** where:

- **Hardware efficiency enables software intelligence:** The reduced energy consumption from thermodynamic optimization makes the device suitable for always-plugged-in, on-demand use, which in turn generates useful operational data.
- **Software intelligence enhances hardware performance:** The mobile app and cloud platform are not afterthoughts but integral components. They provide the user interface for setting schedules (e.g., "run at night when electricity is cheaper"), visualizing long-term usage trends and cost savings, and receiving predictive maintenance alerts that prevent hardware degradation.

This project directly confronts this tripartite gap. It proposes the development of a **holistic household distillation system** that co-designs hardware efficiency (through PID-controlled heating and heat recovery) with software intelligence (through IoT-enabled sensors and a companion mobile application). By bridging the thermodynamic gap, filling the informational void, and achieving true hardware-software integration, this research will deliver a distillation

appliance that is not only energy-efficient but also intelligent, transparent, and seamlessly integrated into the modern connected home fundamentally reimagining what a household distiller can be.

*Table 1. Temperatures of basin water and air, distilled output, and improvement*

| Treatment | Basin water temperature (⁰C) | | Basin air temperature (⁰C) | | Total Average distilled output (mL/ m²/ day) | Improvement compared to the control (T1) (%) |
|---|---|---|---|---|---|---|
| | Maximum | Average | Maximum | Average | | |
| **T1** | 49.2 | 38.8 | 51.6 | 40.8 | 1738.5 | - |
| **T2** | 58.8 | 45.5 | 61.6 | 48.1 | 3006.9 | 72.98 |
| **T3** | 52.6 | 41.1 | 54.0 | 41.4 | 2095.0 | 20.5 |
| **T4** | 49.5 | 46.6 | 55.3 | 42.2 | 2576.3 | 48.2 |
| **T5** | 58.3 | 43.6 | 64.3 | 46.3 | 4143.0 | 138.3 |

*Table 2. Comparison of the total distilled output and output during morning and afternoon of each treatment.*

| Treatment | Average distilled output (mL) | | |
|---|---|---|---|
| | (6.00 am – 12.00 noon) | (12.00 noon – 6.00 pm) | Total output (6.00 am – 6.00 pm) |
| **T1** | $302.4^{C}$ | $1436.1^{D}$ | $1738.5^{E}$ |
| **T2** | $826.4^{B}$ | $2180.6^{B}$ | $3006.9^{B}$ |
| **T3** | $322.5^{C}$ | $1772.5^{C}$ | $2095.0^{D}$ |
| **T4** | $806.9^{B}$ | $1769.4^{C}$ | $2576.3^{C}$ |
| **T5** | $1527.8^{A}$ | $2615.2^{A}$ | $4143.0^{A}$ |

# 2. OBJECTIVE

## 2.1 Main Objective

To design, develop, and validate a **smart, energy-optimized household distilled water generation system** that integrates IoT-based monitoring and mobile application control to enhance user convenience and operational efficiency.

## 2.2 Specific Objectives

1. **Energy-Efficient Distillation Mechanism Design:** To design and construct a distillation unit incorporating passive energy-saving techniques. This includes a well-insulated boiling chamber and, crucially, a **counter-flow heat exchanger** to preheat incoming feed water using the heat of the outgoing distilled water, thereby reducing the energy required to reach boiling point.

2. **IoT-Enabled Control and Monitoring System:** To develop an embedded system using an ESP32 microcontroller to monitor key operational parameters via sensors (temperature of boiling chamber and condenser output, water level in feed and collection tanks, ambient temperature) and to intelligently control the heating element using techniques like PWM or a relay with PID control to maintain optimal temperature and minimize energy use.

3. **Mobile Application Development:** To design and develop a user-friendly mobile application (for both Android and iOS, using a cross-platform framework like Flutter or React Native) that communicates with the ESP32 via Wi-Fi and a cloud backend (e.g., Firebase). The app will display real-time data, allow remote start/stop control, provide energy and production history logs, and send push notifications for user alerts (e.g., "Water tank empty," "Descaling cycle required").

4. **System Integration and Performance Evaluation:** To integrate all hardware and software components into a functional prototype and rigorously evaluate its performance. This includes measuring the system's **energy consumption per liter** of distilled water produced, comparing it to a conventional distiller, and assessing the accuracy of sensor readings, the reliability of the mobile app control, and the overall user experience.

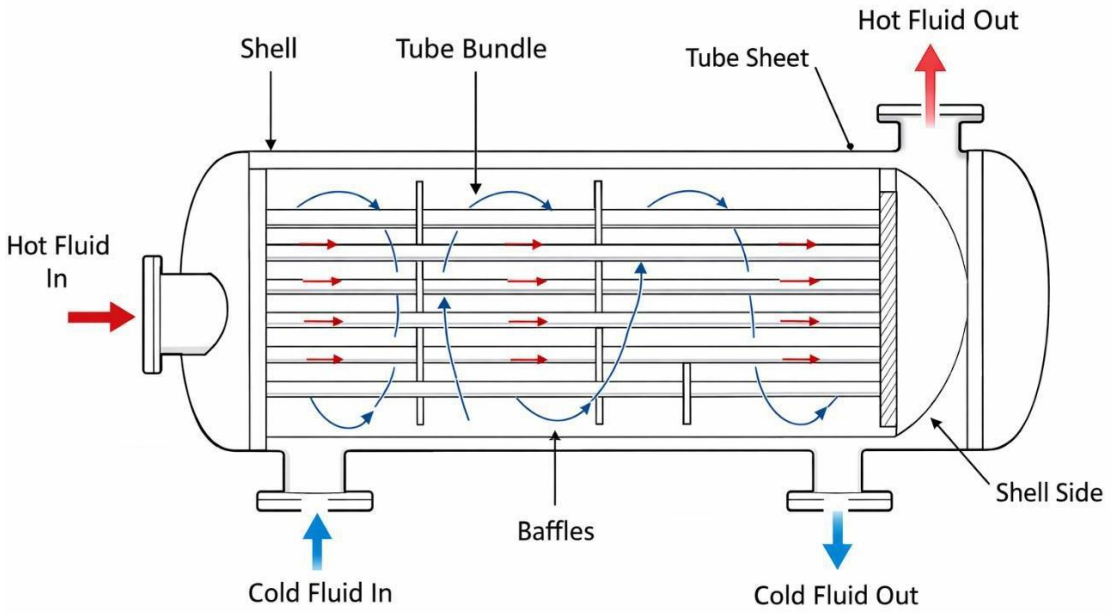



*Figure 3. Schematic Diagram of Heat Exchanger*

## 3. METHODOLOGY

The development will follow a structured engineering design process.

- **Phase 1: Thermodynamic Design and Component Sourcing**
    - **Heat Exchanger Design:** A small, counter-flow tube-in-tube or plate heat exchanger will be designed or sourced. The hot distilled water output will flow through one channel, while cold feed water entering the system will flow through the opposite channel, absorbing heat.
    - **Chamber Design:** A stainless-steel boiling chamber will be used for its corrosion resistance and heat conductivity. High-temperature insulation (e.g., ceramic fiber) will be wrapped around the chamber.
    - **Component Sourcing:** Key components include a 500-750W stainless steel heating element, an insulated water pump for the heat exchanger loop, an ESP32 development board, a solid-state relay (SSR) for controlling the heating element, DS18B20 waterproof temperature sensors, and float switch or ultrasonic water level sensors.

- **Phase 2: Hardware Assembly and Embedded Firmware Development**
    - **Assembly:** The physical prototype will be assembled, integrating the boiling chamber, heat exchanger, condenser coil (passive or fan-assisted), water tanks, and electronics enclosure.
    - **Firmware (Arduino IDE/PlatformIO):** Code will be developed for the ESP32 to:
        - Read temperature sensors via the OneWire protocol.
        - Read water level sensor states.
        - Implement a PID control loop. The PID algorithm will calculate the optimal power output (via PWM to the SSR) to maintain a rolling boil efficiently, rather than simply turning the heater on and off at full power.
        - Control the feed pump for the heat exchanger.
        - Establish a Wi-Fi connection and publish sensor data and system state to a cloud database (e.g., Firebase Realtime Database) using JSON format via HTTP requests or MQTT. It will also listen for incoming commands (start/stop) from the cloud.

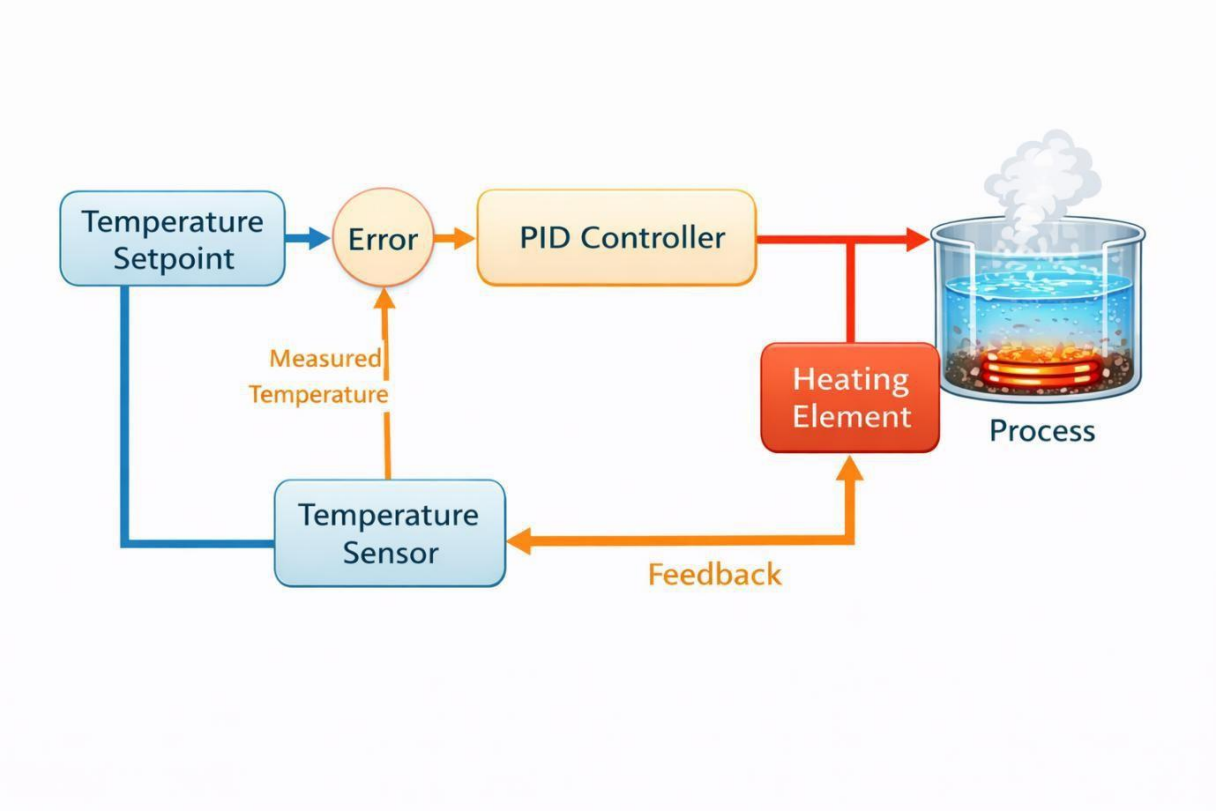



*Figure 4. PID Control Loop for Heating Element*

- **Phase 3: Mobile Application and Cloud Backend Development**
    - **Cloud Setup:** A Firebase project will be created. Firebase Realtime Database or Firestore will store time-stamped sensor data and user settings. Firebase Cloud Messaging (FCM) will be used for push notifications.
    - **Mobile App (Flutter/React Native):** The app will be developed with:
        - **Login/Authentication:** To link users to their specific device.
        - **Real-time Dashboard:** A screen displaying live temperature, water levels, and system status, fetched from the cloud database.
        - **Control Screen:** Buttons to send start/stop commands back to the ESP32 via the cloud.
        - **History & Analytics:** Graphs (e.g., using Flutter's charts_flutter library) showing historical energy consumption and water production per cycle.
        - **Settings & Notifications:** A screen to set user preferences and view push notifications.
- **Phase 4: Integration, Testing, and Validation**
    - **Unit Testing:** Each subsystem (heating, sensing, heat exchanger, app control) will be tested individually.
    - **Efficiency Testing:** The complete system will be run through multiple distillation cycles. The energy consumed (using a plug-in power meter) and the volume of distilled water produced will be recorded to calculate **Wh/L**. This will be compared against a baseline test with the heat exchanger bypassed to quantify energy savings.
    - **Reliability Testing:** The system will be run for extended periods to test for overheating, sensor drifting, and communication stability.
    - **User Acceptance Testing:** A small group of potential users will be asked to use the mobile app and provide feedback on its usability and clarity.

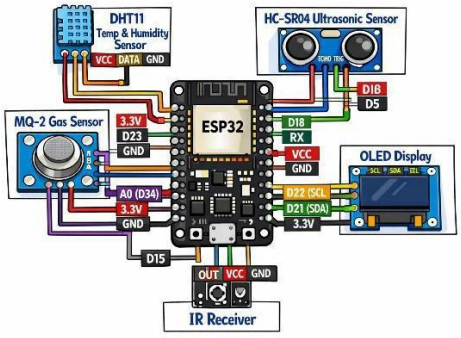



*Figure 5. Circuit Diagram of ESP32 with Sensors*

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

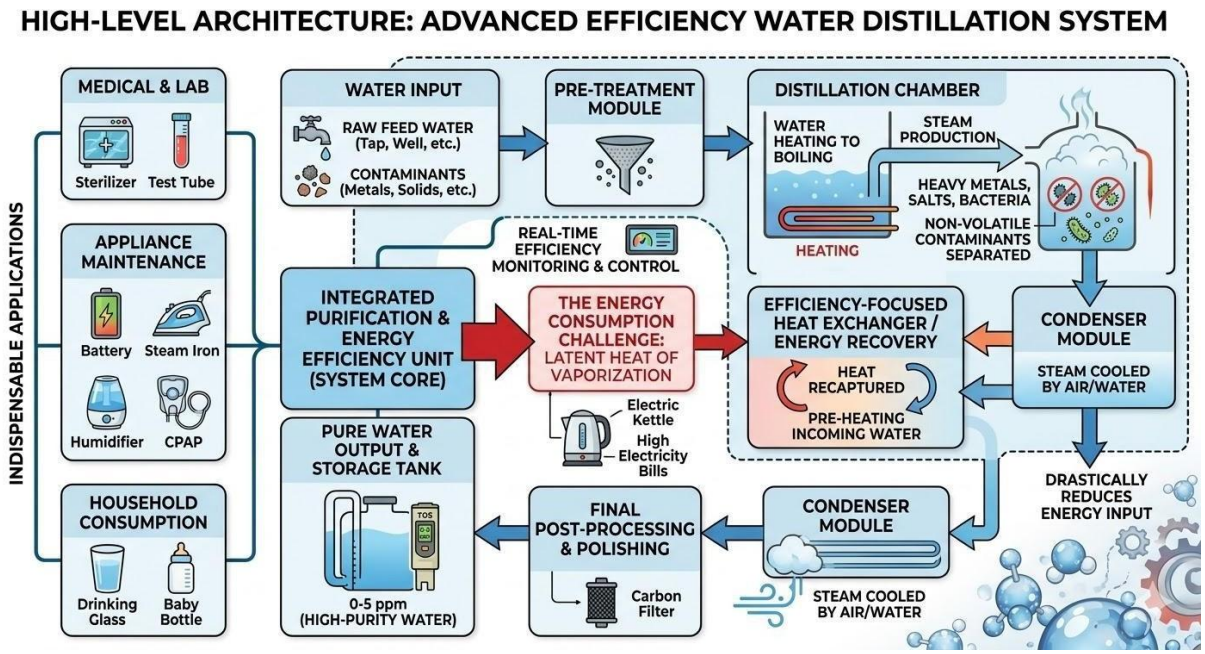



*Figure 6. High-Level System Architecture*

# 5. USER REQUIREMENTS

**Functional Requirements:**

- **Production -** The system shall produce distilled water with a purity level of < 10 ppm TDS.
- **Energy Efficiency -** The system shall achieve a minimum of **30% energy savings** per liter compared to a conventional, non-optimized countertop distiller of similar capacity, primarily through heat recovery and PID control.
- **Monitoring -** The system shall monitor and report the following parameters to the cloud in real-time (every 5-10 seconds): boiling chamber temperature, condenser output temperature, feed water tank level, and distilled water collection tank level.
- **Control -** The user shall be able to remotely start, stop, and schedule distillation cycles via the mobile application.
- **Alerting -** The system shall send push notifications to the user's mobile device for key events, such as "Cycle Complete," "Water Tank Empty," "Collection Tank Full," and "Descaling Recommended."

**Non-Functional Requirements:**

- **Safety -** The system must include multiple safety features: automatic shutdown if the boiling chamber temperature exceeds a safe threshold (e.g., 110°C), and a dry-run cutoff to prevent heating with no water and thermally protected electrical components.
- **Usability -** The mobile application must have an intuitive interface with clear data visualization (gauges, simple icons) and require minimal training to operate.
- **Reliability -** The system should be capable of operating for at least one full distillation cycle (3-4 hours) without any communication dropouts or system failures.
- **Maintainability -** The design should allow for easy access to the boiling chamber for periodic descaling (e.g., with citric acid) and for replacing the feed/collection tanks.
- **Cost-Effectiveness -** The total bill of materials for the prototype should be kept under a target budget (e.g., LKR 30,000-40,000) to ensure potential market viability.

# 6. COMMERCIALIZATION PLAN

This smart distiller has a clear path to market, targeting the growing consumer demand for both health and smart home technology.

- **Target Market Segments:**
    1. **Health-Conscious Homeowners:** Individuals who are concerned about tap water quality and prefer the certainty of distilled water for drinking, cooking, and making beverages like coffee and tea.
    2. **Tech-Savvy Consumers (Smart Home Enthusiasts):** Users who appreciate the convenience of app-connected appliances and are interested in monitoring and optimizing their home energy use.
    3. **Medical & Appliance Users:** Households with members using CPAP machines, requiring water for humidifiers, or wanting to protect expensive steam irons and espresso machines from scale damage. The convenience of on-demand distilled water is a major selling point.
- **Market Strategy:**
    - **Online Channels:** Launch via e-commerce platforms (Daraz, Amazon) and a dedicated website, leveraging social media marketing (Facebook, Instagram) to demonstrate the product's benefits, especially energy savings and app control.
    - **Partnerships:** Collaborate with retailers of medical supplies (for CPAP users) and high-end appliance stores to position the distiller as an essential accessory.
- **Revenue Model:**
    - **Product Sales:** Primary revenue from the sale of the hardware unit.
    - **Premium App Features:** A freemium model where basic monitoring is free, but advanced features like detailed energy analytics, usage history export, and customizable automation schedules are available via a one-time purchase or low-cost subscription.
    - **Consumables:** Revenue from selling replacement parts, such as replacement water tanks, cleaning/descaling cartridges, or replacement sensors.
- **Competitive Advantage:** The unique selling point is the **combination of significant energy efficiency and full smart home integration**. While simple distillers exist, none offer real-time energy monitoring, PID-optimized heating for efficiency, remote control, and proactive maintenance alerts in a single, affordable package. This positions the product as the premium, intelligent choice in a niche market.

# 7. BUDGET AND JUSTIFICATION

*Table 3. Budget and Justification*

| Item | Component/Description | Estimated Cost (LKR) | Justification |
|---|---|---|---|
| **1. Distillation Core** | Stainless Steel Boiling Chamber, 750W Heating Element, High-Temp Insulation | 3,500 | The heart of the system; material choice and insulation are key for efficiency and safety. |
| **3. Condenser Unit** | Copper or Stainless-Steel Condenser Coil, Small Fan (optional) | 8,000 | For efficiently cooling and condensing the steam. |
| **4. Enclosure & Tanks** | Outer Housing (Plastic or Metal), BPA-free Feed and Collection Tanks (2x) | 4,000 | For structural integrity, aesthetics, and safe water storage. |
| **5. Core Electronics** | ESP32 Development Board, Power Supply (12V/5V), Solid State Relay (SSR) | 5,500 | The "brain" of the system is for control, connectivity, and safe switching of high power. |
| **7. Misc. Hardware** | Tubing, Fittings, Waterproof Connectors, Wires, Screws, etc. | 4,500 | Essential for plumbing and assembling all components. |
| **TOTAL ESTIMATED BUDGET** | | **25,500** | |

## 8. WORK BREAKDOWN STRUCTURE (WBS)




*Figure 7. Work Breakdown Structure (WBS)*

# 9. GANTT CHART

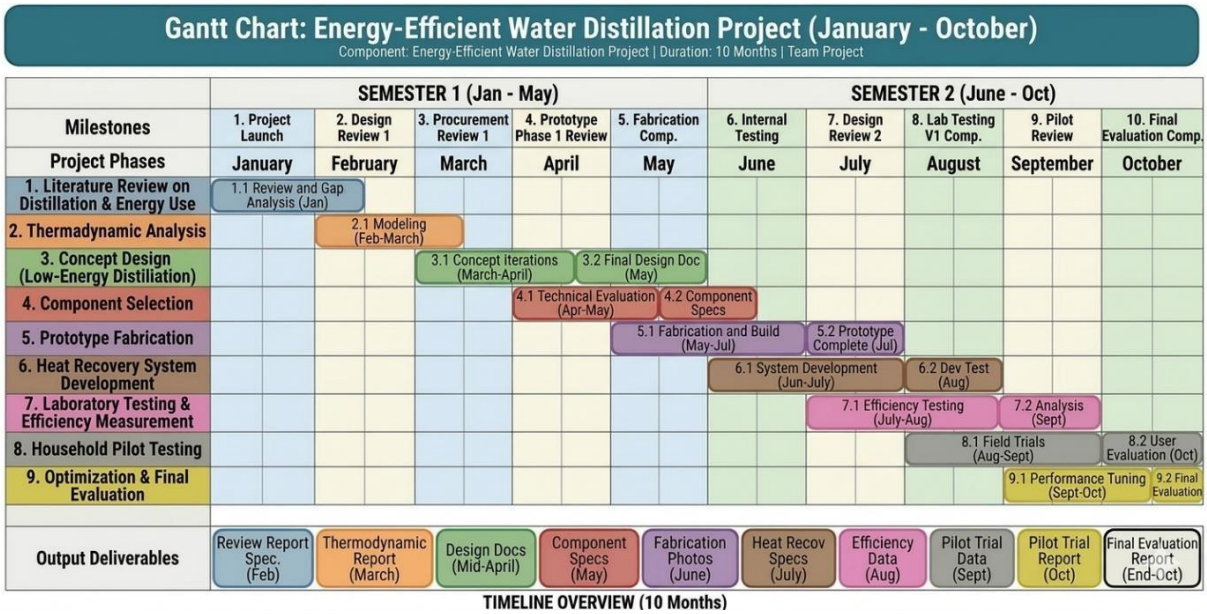



*Figure 8. Gantt Chart*

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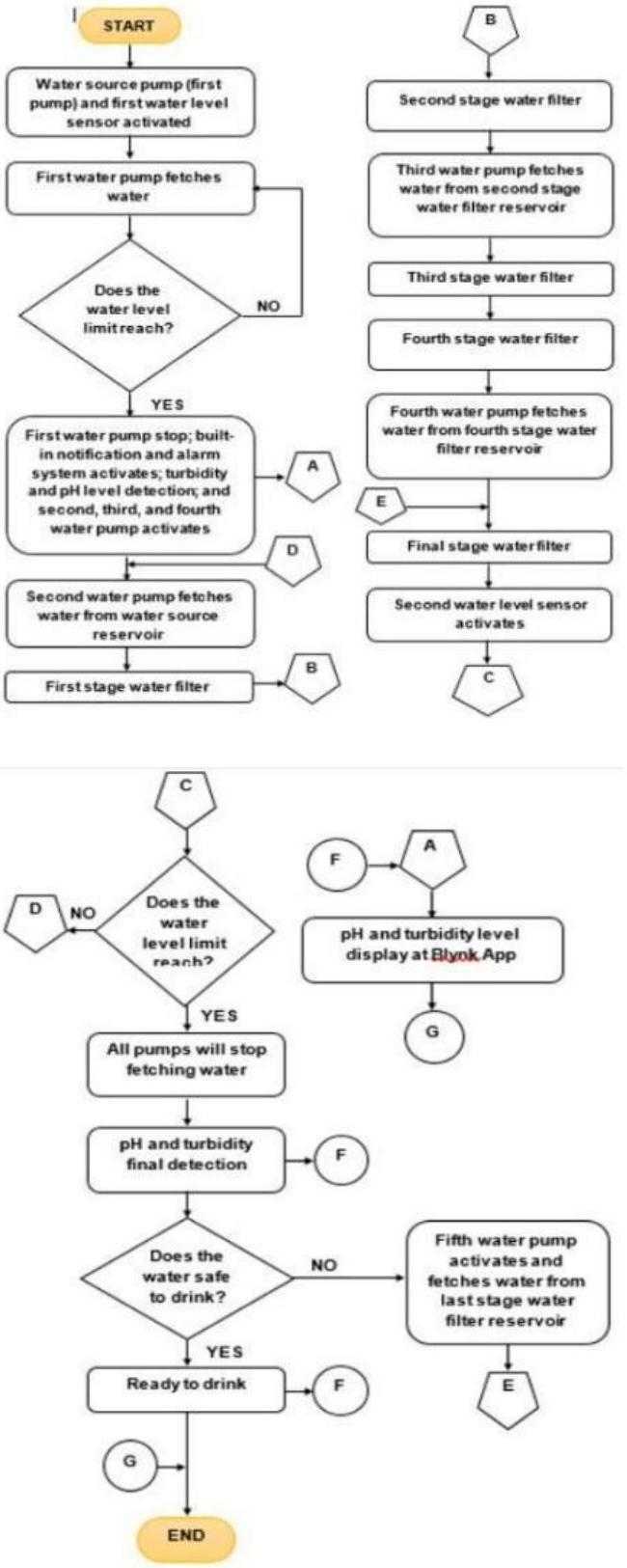
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# 11. APPENDICES




*Appendix A. System Flow Chart*

- It presents the detailed works of the filtration machine. It explains the whole process of how the project operates. It is composed of sensors that can detect 3 major inputs which are water level sensor, pH sensor, and turbidity sensor. The water level sensor triggers the pumping system of the project. The pH level sensor detects the acidity or alkalinity of the water. Lastly, turbidity sensors detect the cloudiness or haziness of the water. When the input sensors will activate the built-in notification and alarm system of the device.




*Appendix B. Workflow of the water quality monitoring system*

- This diagram shows a smart water monitoring system where sensor data is collected by a Raspberry Pi, preprocessed, and stored locally. The data is then classified into three water quality levels based on physicochemical standards, with each category analyzed using machine learning models for anomaly detection. The system also compares results with laboratory data to ensure reliable and accurate water quality assessment.

| pH | Turbidity | Temperature | TDS | DateTime | Quality | Label |
|---|---|---|---|---|---|---|
| 7.32 | 4 | 19.47 | 615 | 2024-07-01 08:00:00 | 0.5 | |
| 7.52 | 4 | 18.81 | 600 | 2024-07-01 08:00:00 | 0.5 | |
| 7.46 | 4 | 18.92 | 600 | 2024-07-01 08:00:00 | 0.5 | |
| 7.73 | 3.5 | 20.15 | 515 | 2024-07-01 08:00:00 | 0.5 | |
| 7.56 | 3.9 | 24.82 | 516 | 2024-07-01 08:00:00 | 0.5 | |
| 7.59 | 4 | 27.34 | 555 | 2024-07-01 14:00:00 | 0.5 | |
| 7.44 | 3.9 | 29.32 | 700 | 2024-07-01 14:00:00 | 0.5 | |
| 7.23 | 4.1 | 31.28 | 500 | 2024-07-01 14:00:00 | 0.5 | |
| 7.33 | 2.4 | 30.88 | 516 | 2024-07-02 14:00:00 | 0.5 | |
| 7.33 | 4 | 30.26 | 511 | 2024-07-03 14:00:00 | 0.5 | |
| 7.21 | 3.5 | 27.09 | 511 | 2024-07-04 14:00:00 | 0.5 | |
| 7.4 | 3.5 | 22.98 | 500 | 2024-07-01 14:00:00 | 0.5 | |
| 7.21 | 4 | 21.55 | 559 | 2024-07-01 14:00:00 | 0.5 | |
| 7.51 | 3.5 | 19.05 | 505 | 2024-07-01 14:00:00 | 0.5 | |
| 7.82 | 2.4 | 16.98 | 700 | 2024-07-01 14:00:00 | 0.5 | |
| 7.65 | 4 | 17.95 | 600 | 2024-07-01 14:00:00 | 0.5 | |
| 7.69 | 3.5 | 24.41 | 500 | 2024-07-01 14:00:00 | 0.5 | |
| 7.42 | 4 | 26.78 | 600 | 2024-07-01 14:00:00 | 0.5 | |

*Appendix C. labels based on the physicochemical parameters of water quality*

- The dataset used in the analysis is categorized into three classification labels based on water quality physicochemical parameters, with the majority (69%) falling under the "Borderline Condition" category. About 25% of the samples "Meet Physicochemical Criteria," while only 6% are classified as "Does Not Meet Physicochemical Criteria," indicating that most water samples are of borderline quality, with a small proportion being clearly non-compliant.